Deep Reinforcement Learning with Dynamic Action Space for QoE-Driven Scheduling in MIMO-OFDM Wi-Fi Networks

Yulan Yuan[1], Huayan Guo[2], Danny H. K. Tsang[1], Vincent K. N. Lau[3]
[1]The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China
[2]The Hong Kong University of Science and Technology Shenzhen Research Institute, Shenzhen, China
[3]The Hong Kong University of Science and Technology, Hong Kong, China
Email: yyuan202@connect.hkust-gz.edu.cn, eeguohuayan@ust.hk, eetsang@ust.hk, eeknlau@ece.ust.hk

***Abstract*—This paper addresses QoE-driven scheduling in single-AP multi-STA MIMO-OFDM Wi-Fi networks. Traditional CSI-based schedulers ignore application-layer QoE metrics, while existing DRL approaches suffer from high complexity in large action spaces. We propose a low-complexity DNN-DRL framework that first uses a pre-trained DNN to select candidate STA pairs based on CSI, then employs a DRL agent to make final scheduling decisions by jointly optimizing CSI and QSI. Formulated as a Markov Decision Process with multi-metric QoE rewards, the proposed method significantly reduces playback interruption and buffer overflow, achieving superior QoE and efficiency with low complexity.**

***Index Terms*—Wi-Fi Scheduling, QoE, MU-MIMO, OFDM, Reinforcement Learning**

I. Introduction

The increasing demand for high-quality video services such as 4K/8K, AR, and VR imposes stringent Quality of Experience (QoE) requirements on wireless networks [1]. Unlike traditional data traffic, video streaming is highly sensitive to playback interruption and buffer overflow [2], [3], which lead to video freezing and wasted throughput, respectively. Therefore, an effective scheduling framework must jointly account for these QoE metrics while adapting to dynamic channel and queue states.

Existing wireless scheduling methods fall into conventional optimization-based and learning-based approaches. Conventional methods typically exploit channel state information (CSI) to maximize throughput or fairness [4], [5]. While heuristic schemes offer low complexity, they underutilize CSI, whereas greedy or exhaustive methods [6], [7] suffer from prohibitive complexity as the number of users increases. User clustering [8], [9] reduces complexity but static grouping often leads to performance loss. Moreover, CSI-only designs neglect queue state information (QSI), which is critical for video streaming due to buffer dynamics. MDP-based formulations incorporating both CSI and QSI [10], [11] are effective for small systems but become intractable in large-scale scenarios.

Deep reinforcement learning (DRL) alleviates MDP scalability issues by approximating policies with deep neural networks [12]–[14] and has shown strong performance in dynamic wireless environments. However, large action spaces in multiuser MIMO scheduling remain challenging. Existing solutions [15], [16] reduce complexity via partitioning or scoring but rely on predefined structures or complex networks, limiting performance or increasing computation. Additionally, high-dimensional CSI in wideband MIMO-OFDM systems further complicates training. Thus, efficient DRL frameworks that jointly exploit CSI and QSI while handling large state and action spaces are still needed.

To address these challenges, we propose a low-complexity DRL framework for QoE-driven user scheduling and power allocation in Wi-Fi networks. A hierarchical DNN architecture is introduced to efficiently manage high-dimensional states and large decision spaces under dynamic traffic and channel conditions. Our main contributions are summarized as follows: **1) QoE-Aware MDP Formulation:** An MDP that jointly incorporates CSI and QSI, with a reward capturing playback interruption, buffer overflow, delay, and power consumption. **2) Low-Complexity DRL Framework:** A hierarchical design combining a CSI-based user scoring module and a QoE-aware policy network for joint scheduling and power control. **3) Performance Gains:** Simulation results show improved QoE and resource efficiency over state-of-the-art baselines with low computational complexity.

II. System Model and Problem Formulation

A. System Model

Consider an AP with N_t antennas serving U STAs, $\mathcal{U} = \{1, \ldots, U\}$, each equipped with N_r antennas and receiving

This work was supported in part by the National Natural Science Foundation of China under Grant 62101472, in part by the Research Grants Council (RGC), Hong Kong, under Project 16207221, in part by the Research Grants Council (RGC), Hong Kong, under the Areas of Excellence Scheme Grant AoE/E-101/23-N and AoE/E-601/22-R, in part by the National Foreign Expert Project under Project H20240994, in part by Guangdong Provincial Project under Grant 2021JC02X149, in part by Guangzhou Municipal Science and Technology Project under Grant 2023A03J0011, in part by Guangzhou Municipal Key Laboratory on Future Networked Systems (024A03J0623), and in part by Guangdong Provincial Key Laboratory of Integrated Communications, Sensing and Computation for Ubiquitous Internet of Things (No.2023B1212010007).

a single data stream. Up to $N_\mathrm{u} \le N_\mathrm{t}$ users can be spatially multiplexed. The application type of STA u is denoted by $\alpha_u \in \{0,1\}$, where $\alpha_u = 0$ and $\alpha_u = 1$ represent video streaming and data traffic, respectively. Time is slotted with duration τ, and $\beta_u(t) \in \{0,1\}$ indicates whether STA u is scheduled at slot t. The system employs N subcarriers, and the channel from the AP to STA u on subcarrier n is $\mathbf{H}_{u,n}(t) \in \mathbb{C}^{N_\mathrm{r}\times N_\mathrm{t}}$, assumed i.i.d. over time. The received signal is

$$\begin{aligned}\mathbf{y}_{u,n}(t) = \beta_u(t)\mathbf{H}_{u,n}(t)\mathbf{f}_{u,n}(t)s_{u,n}(t) \\ +\sum_{v\neq u}\beta_v(t)\mathbf{H}_{u,n}(t)\mathbf{f}_{v,n}(t)s_{v,n}(t) + \mathbf{z}_{u,n}(t),\end{aligned} \tag{1}$$

where $\mathbf{f}_{u,n}(t)$ is the precoding vector, $s_{u,n}(t)$ the transmit symbol, and $\mathbf{z}_{u,n}(t) \sim \mathcal{CN}(0, \nu_z^2\mathbf{I})$ the noise. To model heterogeneous traffic, each video STA maintains a playback buffer, while the AP holds a data buffer for data traffic [2], [3]. For video streaming ($\alpha_u = 0$), the buffer occupancy $Q_u(t)$ is bounded by capacity N_u, with interruptions and overflows occurring at $Q_u(t) = 0$ and buffer saturation, respectively. For data traffic ($\alpha_u = 1$), the AP buffer evolves as a queue with random arrivals $A_u(t)$ of mean rate $\lambda_u = \mathbb{E}_t[A_u(t)]$, yielding an average delay $\mathbb{E}_t[Q_u(t)]/\lambda_u$ [17]. Let $\mathcal{K}(t) = \{u \mid \beta_u(t) = 1\}$ denote the scheduled user set. Using SVD-based zero-forcing precoding, the dominant right singular vectors form $\mathbf{V}_n(t) = [\bar{\mathbf{v}}_{u,n}(t)]_{u\in\mathcal{K}(t)} \in \mathbb{C}^{N_\mathrm{t}\times N_\mathrm{u}}$. The transmit precoder is

$$\mathbf{F}_n(t) = \eta_n(t)\mathbf{V}_n(t)\big(\mathbf{V}_n^\mathrm{H}(t)\mathbf{V}_n(t)\big)^{-1}\operatorname{diag}\big(\sqrt{P_u(t)}\big)_{u\in\mathcal{K}(t)}, \tag{2}$$

where $P_u(t)$ is the transmit power of user u, and $\eta_n(t) = \frac{1}{\left\|\mathbf{V}_n(t)(\mathbf{V}_n^\mathrm{H}(t)\mathbf{V}_n(t))^{-1}\right\|_F}$. ensures power normalization. The resulting SNR and achievable rate are

$$\mathrm{SNR}_{u,n}(t) = \frac{\eta_n^2(t)P_u(t)\sigma_{u,n}^2(t)}{\nu_z^2}, \tag{3}$$

$$R_u(t) = \beta_u(t)\sum_{n=1}^{N}\log_2(1 + \mathrm{SNR}_{u,n}(t)), \tag{4}$$

where $P_u(t)$ directly controls the SNR and achievable rate.

B. Problem Formulation

At each slot t, the controller observes the system state $\boldsymbol{\chi}(t) = (\mathcal{H}(t), \mathcal{Q}(t))$, where $\mathcal{H}(t) = \big\{\mathbf{H}_{u,n}(t)\,\big|\, u \in \mathcal{U},\ n = 1,\ldots,N\big\}$, $\mathcal{Q}(t) = \big\{Q_u(t)\,\big|\, u \in \mathcal{U}\big\}$, and then selects an action $\boldsymbol{y}(t) = \{\mathcal{M}(t), \mathbf{P}(t)\}$, where $\mathcal{M}(t)$ denotes the scheduled user set and $\mathbf{P}(t) = [P_1(t),\ldots,P_U(t)]$ represents the power allocation. The constraints are

$$|\mathcal{M}(t)| = N_\mathrm{u}, \quad \sum_{u\in\mathcal{K}(t)} P_u(t) \le P_\mathrm{M}, \tag{5}$$

where P_M is the maximum transmit power of the AP. For data users ($\alpha_u = 1$), the queue dynamics are

$$Q_u(t+1) = [Q_u(t) - R_u(t)\tau]^+ + A_u(t)\tau, \tag{6}$$

while for video users ($\alpha_u = 0$),

$$Q_u(t+1) = \min\{[Q_u(t) - \mu_u\tau]^+ + R_u(t)\tau, W_u\}, \tag{7}$$

where μ_u and W_u denote the streaming rate and buffer capacity.

1) Long-Term Cost with QoE Metrics: We jointly consider three QoE-related metrics: delay, power consumption, and playback smoothness. Under a given policy π, the instantaneous delay for data STAs is defined as $D_u(t) = Q_u(t)/\lambda_u$, and the instantaneous playback deviation for video STAs is $V_u(t) = (Q_u(t) - Q_u^*)^2$, where $Q_u^* = \rho W_u$ represents the desired playback level with $\rho \in (0,1)$ controlling the steady-state buffer ratio (typically $\rho = 0.5$). The corresponding long-term averages under policy π are given by $\bar{D}_u(\pi) = \alpha_u\,\mathbb{E}_t^\pi[D_u(t)], \bar{P}_u(\pi) = \mathbb{E}_t^\pi[P_u(t)], \bar{V}_u(\pi) = (1 - \alpha_u)\,\mathbb{E}_t^\pi[V_u(t)]$, where the expectation is taken over trajectories induced by sampling actions $\boldsymbol{y}(t) \sim \pi(\cdot \mid \boldsymbol{\chi}(t))$. Here, $\bar{D}_u(\pi)$ reflects average delay for data applications, $\bar{P}_u(\pi)$ captures long-term power consumption, and $\bar{V}_u(\pi)$ quantifies playback smoothness for video applications.

2) Optimization Problem: The goal is to find the policy π^* minimizing the weighted sum of QoE costs:

$$\textbf{(P1):}\quad \min_{\pi}\ \zeta(\pi) = \sum_{u\in\mathcal{U}}\big[\omega_\mathrm{d}\bar{D}_u(\pi) + \omega_\mathrm{v}\bar{V}_u(\pi) + \omega_\mathrm{p}\bar{P}_u(\pi)\big], \quad \text{s.t.}\quad (5), \tag{8}$$

3) Challenges: Solving **(P1)** is challenging due to the combinatorial action space for user scheduling, $\binom{U}{N_\mathrm{u}}$ combinations, and the high-dimensional state $\mathcal{H}(t)$ spanning $U \times N \times N_\mathrm{t} \times N_\mathrm{r}$ parameters per slot. Representing the policy π is also difficult: classical Q-tables fail for continuous queue states $\mathcal{Q}(t)$, and DRL methods like DQN struggle with continuous or coupled actions [2], [12], [13]. A policy-gradient approach is adopted to directly parameterize π, though training remains challenging due to the large combinatorial output space.

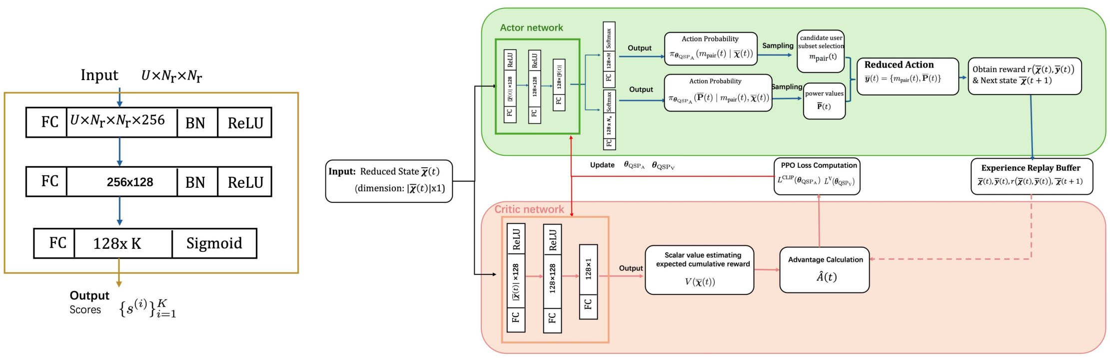
III. Low-Complexity Solution with Efficient User Pairing and Power Control

A. Framework Overview

Directly applying DRL to (**P1**) is computationally prohibitive due to the large state–action space. To address this, we propose a two-stage hierarchical framework. **CSI-based Scoring Network (CSIS-Net)** leverages channel covariance matrices instead of instantaneous CSI to reduce input dimensionality and outputs pairing scores for candidate STA subsets, yielding a reduced and adaptive action space. With CSIS-Net fixed, the **QoE-Aware Scheduling Policy Network (QSP-Net)** employs a PPO-based DRL agent that observes the reduced state composed of queue information and top candidate subsets, and learns joint scheduling and power allocation to maximize long-term QoE under (**P1**).

B. Pre-Training for CSI-Based Scoring Network (CSIS-Net)

1) Input Representation and Feature Processing: CSIS-Net is designed to extract key spatial features from high-dimensional MIMO-OFDM channels. Instead of using the full channel tensor $\mathcal{H} \in \mathbb{C}^{U\times N\times N_\mathrm{t}\times N_\mathrm{r}}$, it takes the receive-side covariance matrices as input: $\mathbf{C}_u = \frac{1}{N}\sum_{n=1}^{N}\mathbf{H}_{u,n}\mathbf{H}_{u,n}^H, \quad u \in \mathcal{U}$, where $\mathbf{C}_u \in \mathbb{C}^{N_\mathrm{r}\times N_\mathrm{r}}$ captures the spatial characteristics



(a) Architecture of CSIS-Net

(b) Architecture of QSP-Net

Fig. 1. Architecture of the proposed CSIS-Net and QSP-Net. Note that "FC $a \times b$" denotes a fully-connected layer with a input neurons and b output neurons and "BN" denotes a batch normalization layer.

of user u while significantly reducing input dimensionality. CSIS-Net processes the covariance set $\mathcal{C} \triangleq \{\mathbf{C}_u \mid u \in \mathcal{U}\}$ and outputs quality scores for all user subsets of size N_{u}. Let $\mathfrak{K} = \left\{\tilde{\mathcal{K}}^{(1)}, \tilde{\mathcal{K}}^{(2)}, \ldots, \tilde{\mathcal{K}}^{(K)}\right\}$, where $K = \binom{|\mathcal{U}|}{N_{\mathrm{u}}}$ and $\tilde{\mathcal{K}}^{(i)} = \{v_1^{(i)}, \ldots, v_{N_{\mathrm{u}}}^{(i)}\}$ denotes the ith candidate subset. The corresponding scores are given by $\{s^{(i)}\}_{i=1}^{K} = f_{\mathrm{p}}(\mathcal{C}; \boldsymbol{\theta}_{\mathrm{CSIS}})$, where $f_{\mathrm{p}}(\cdot; \boldsymbol{\theta}_{\mathrm{CSIS}})$ denotes the CSIS-Net mapping and $s^{(i)}$ is the score of $\tilde{\mathcal{K}}^{(i)}$.

2) Training Loss Function: CSIS-Net is trained to prioritize user subsets with stronger channel conditions, thereby maximizing the expected throughput under equal power allocation. The quality of each candidate subset $\tilde{\mathcal{K}}^{(i)}$ is quantified by its average received signal power: $\xi^{(i)} = \frac{1}{N} \sum_{n=1}^{N} \frac{1}{\mathrm{Tr}\left[\left(\mathbf{H}_n(\tilde{\mathcal{K}}^{(i)}) \mathbf{H}_n^H(\tilde{\mathcal{K}}^{(i)})\right)^{-1}\right]}$, where $\mathbf{H}_n(\tilde{\mathcal{K}}^{(i)}) = \left[\mathbf{H}_{v_1^{(i)}, n}, \mathbf{H}_{v_2^{(i)}, n}, \ldots, \mathbf{H}_{v_{N_{\mathrm{u}}}^{(i)}, n}\right]$ denotes the aggregate channel matrix of subset $\tilde{\mathcal{K}}^{(i)}$ on subcarrier n. The loss function is formulated as $\mathcal{L}_{\mathrm{CSIS}}(\boldsymbol{\theta}_{\mathrm{CSIS}}; \mathcal{C}, \mathcal{H}) = -\sum_{i=1}^{K} \xi^{(i)} s^{(i)} + \lambda \sum_{i=1}^{K} s^{(i)}$, where λ is a regularization coefficient. The first term encourages higher scores for subsets with better channel quality $\xi^{(i)}$, while the regularization term prevents trivial large-value solutions by penalizing excessive score magnitudes.

3) Training Details: The training objective of CSIS-Net is then defined as

$$\textbf{(P2)} : \min_{\boldsymbol{\theta}_{\mathrm{CSIS}}} \mathbb{E}[\mathcal{L}_{\mathrm{CSIS}}(\boldsymbol{\theta}_{\mathrm{CSIS}}; \mathcal{C}, \mathcal{H})] \,. \tag{9}$$

The model is trained using the Adam optimizer with an initial learning rate of 10^{-3} and momentum parameters $(0.9, 0.999)$. The training data are generated from the IEEE 802.11ax TGax channel model (Model-B delay profile) over diverse propagation environments and transmitter–receiver distances. The CSIS-Net architecture is shown in Fig. 1(a).

4) Final Output Message from CSIS-Net to QSP-Net: After optimization, the CSIS-Net parameters are fixed as $\boldsymbol{\theta}^{\star}_{\mathrm{CSIS}}$. Given the covariance set $\mathcal{C}$, CSIS-Net outputs the top-M candidate STA subsets: $\boldsymbol{\phi}(\boldsymbol{\theta}^{\star}_{\mathrm{CSIS}}) = \left\{\left(\tilde{\mathcal{K}}^{(m)}, \xi^{(m)}\right)\right\}_{m=1}^{M}$, where $\tilde{\mathcal{K}}^{(m)} = \{v_1^{(m)}, \ldots, v_{N_{\mathrm{u}}}^{(m)}\}$ and $\xi^{(m)}$ denote the mth subset and its score, respectively. These ranked subsets form the reduced input to QSP-Net.

C. QoE-Aware Scheduling Policy Network (QSP-Net)

1) Reduced State and Action Space: With CSIS-Net optimized at $\boldsymbol{\theta}^{\star}_{\mathrm{CSIS}}$, the reduced state is defined as $\overline{\boldsymbol{\chi}}(t) = \{\mathcal{Q}(t), \boldsymbol{\phi}(t; \boldsymbol{\theta}^{\star}_{\mathrm{CSIS}})\}$, where $\mathcal{Q}(t)$ denotes the queue state and $\boldsymbol{\phi}(t; \boldsymbol{\theta}^{\star}_{\mathrm{CSIS}})$ represents the top-M candidate user subsets and their quality metrics. Based on this reduced state, QSP-Net explores a compact action space including a candidate subset selection and a power allocation vector. At slot t, the agent selects $m_{\mathrm{pair}}(t) \in \{1, 2, \ldots, M\}$ and determines $\overline{\mathbf{P}}(t) = [\overline{P}_1(t), \ldots, \overline{P}_{N_{\mathrm{u}}}(t)]$, where $\overline{P}_i(t)$ is the transmit power of the ith user in $\tilde{\mathcal{K}}^{(m_{\mathrm{pair}}(t))}$. The reduced action is thus $\overline{\boldsymbol{y}}(t) = \{m_{\mathrm{pair}}(t), \overline{\mathbf{P}}(t)\}$.

2) Network Input, Output, and Correspondence to System-Level Actions: The input to QSP-Net is the reduced state $\overline{\boldsymbol{\chi}}(t)$. QSP-Net consists of an *actor network* with parameters $\boldsymbol{\theta}_{\mathrm{QSP_A}}$ and a *critic network* with parameters $\boldsymbol{\theta}_{\mathrm{QSP_V}}$, where $\boldsymbol{\theta}_{\mathrm{QSP}} = \{\boldsymbol{\theta}_{\mathrm{QSP_A}}, \boldsymbol{\theta}_{\mathrm{QSP_V}}\}$. Given $\overline{\boldsymbol{\chi}}(t)$, the actor outputs the joint policy $\pi_{\boldsymbol{\theta}_{\mathrm{QSP_A}}}(\overline{\boldsymbol{y}}(t) \mid \overline{\boldsymbol{\chi}}(t)) = \pi_{\boldsymbol{\theta}_{\mathrm{QSP_A}}}(m_{\mathrm{pair}}(t) \mid \overline{\boldsymbol{\chi}}(t)) \, \pi_{\boldsymbol{\theta}_{\mathrm{QSP_A}}}(\overline{\mathbf{P}}(t) \mid m_{\mathrm{pair}}(t), \overline{\boldsymbol{\chi}}(t))$, where the first term denotes the categorical distribution over M candidate subsets and the second specifies the conditional power allocation. The reduced action $\overline{\boldsymbol{y}}(t)$ is mapped to the original scheduling variables by selecting the user set $\mathcal{K}(t) = \tilde{\mathcal{K}}^{(m_{\mathrm{pair}}(t))}$ and reconstructing the transmit power vector $\mathbf{P}(t)$ such that each user $u \in \mathcal{K}(t)$ is assigned the corresponding power $\overline{P}_i(t)$ if $u = v_i^{(m_{\mathrm{pair}}(t))}$, while all other users are allocated zero power, where $\{v_i^{(m_{\mathrm{pair}}(t))}\}$ denotes the indices of users in the selected subset. The scheduling indicators are recovered by $\beta_u(t) = 1$ if $u \in \mathcal{K}(t)$ and $\beta_u(t) = 0$ otherwise, ensuring a one-to-one mapping between the reduced and original action spaces.

3) Reward and Policy Optimization: The instantaneous reward for action $\overline{\boldsymbol{y}}(t)$ is $r(\overline{\boldsymbol{\chi}}(t), \overline{\boldsymbol{y}}(t)) = -\sum_{u\in\mathcal{U}} \Big(\omega_{\rm d} D_u(t) + \omega_{\rm v} V_u(t) + \omega_{\rm p} P_u(t)\Big)$, with $D_u(t)$, $V_u(t)$, and $P_u(t)$ denoting delay, playback deviation, and power consumption. The training objective is to maximize the expected discounted reward:

$$\textbf{(P3)}: \quad \max_{\boldsymbol{\theta}_{\rm QSP}} J(\boldsymbol{\theta}_{\rm QSP}) = \mathbb{E}\left[\sum_{t=0}^{\infty} \zeta^t \, r\big(\overline{\boldsymbol{\chi}}(t), \overline{\boldsymbol{y}}(t)\big)\right], \quad \text{s.t.} \quad (5), \tag{10}$$

where $\zeta \in (0,1)$ is the discount factor. QSP-Net employs PPO to jointly update $\boldsymbol{\theta}_{\rm QSP}$. The overall QSP-Net architecture is shown in Fig. 1(b).

D. Complexity Reduction Analysis

1) State Space Reduction: The original RL state $\boldsymbol{\chi}(t) = (\mathcal{H}(t), \mathcal{Q}(t))$ includes the full CSI matrix $\mathcal{H}(t)$ and queue states $\mathcal{Q}(t)$, where $|\mathcal{H}(t)| = U \cdot N \cdot N_{\rm t} \cdot N_{\rm r}$ grows rapidly with the number of users U and subcarriers N. Using the hierarchical approach, the reduced state has dimension $|\overline{\boldsymbol{\chi}}(t)| = U + M \cdot (1 + N_{\rm u})$, where the second term comes from $\boldsymbol{\phi}(t; \boldsymbol{\theta}^{\star}_{\rm CSIS})$, achieving a reduction factor of $O\big(U \cdot N \cdot N_{\rm t} \cdot N_{\rm r} / [U + M(1+N_{\rm u})]\big)$.

2) Action Space Reduction: Evaluating all user subsets $\mathcal{K}$ requires combinatorial complexity $D_{\rm dim} = \binom{U}{K}$, which grows exponentially with U. By selecting only the Top-M candidate subsets, the effective action space is reduced to $D_{\rm reduced} = O(M)$, yielding a reduction factor of $O\big(\binom{U}{K}/M\big)$.

3) Training Complexity: The per-iteration complexity is $O(E \cdot T \cdot |\overline{\boldsymbol{\chi}}| \cdot D_{\rm reduced} \cdot D_{\rm dep})$, where E is the number of episodes, T the episode length, and $D_{\rm dep}$ the network depth. The combined reduction in state and action dimensions improves training efficiency compared to conventional DRL.

IV. Simulation Experiments

A. Simulation Setup

The system adopts IEEE 802.11ax MIMO-OFDM with a 160 MHz bandwidth, using $N_{\rm t} = 4$ transmit and $N_{\rm r} = 2$ receive antennas to schedule two users per slot over the TGax Channel Model B at 5 GHz. Each user has a buffer size of $W_u = 100$ Mbits and a slot duration of $\tau = 0.1$ s, with transmit power ranging from 1 to 27 dBm. Channel realizations are generated using the MATLAB WLAN Toolbox for users located 3–15 m from the AP. A single AP serves eight STAs, including four video users (two high-rate users with $\mu_u = 540$ Mbps and two moderate-rate users with $\mu_u = 200$ Mbps) and four data users with $A_u = 20$ Mbps. The proposed method selects the top-$M = 3$ candidate user subsets for scheduling, and four baseline schemes are considered for comparison: **Baseline 1 (Round-Robin):** Users served sequentially, ignoring CSI/QSI;. **Baseline 2 (Maximum Rate):** User with highest instantaneous rate selected, ignoring QSI; **Baseline 3 (Pre-Group DRL) [15]:** Users pre-divided into groups; agent schedules only within the same group; **Baseline 4 (Full-Action DRL) [16]:** Uses implicit DNN to handle full action space without shrinking.

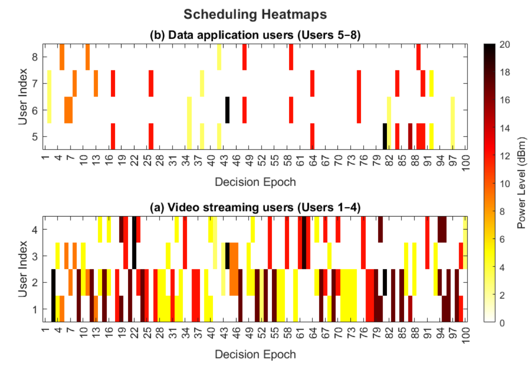


Fig. 2. Power allocation under the proposed algorithm. (a) Video streaming users (Users 1–4). (b) Data application users (Users 5–8).

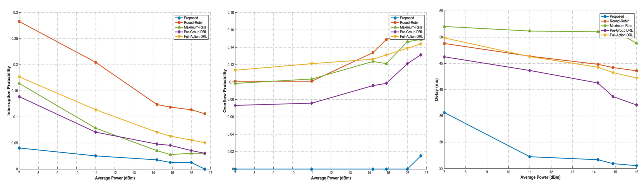


Fig. 3. Performance comparison: (a) Playback interruption probability; (b) Playback overflow probability; (c) Delay.

B. Performance Evaluation

Fig. 2 illustrates the power allocation under the proposed algorithm. In Fig. 2(a), video users 1–4 exhibit differentiated scheduling: high-rate users (1–2) are scheduled more frequently and allocated higher power to avoid playback interruptions, while moderate-rate users (3–4) are scheduled less often but still maintain stable transmission. Fig. 2(b) shows that power allocation among data users 5–8 is relatively balanced and overall lower, indicating that the proposed strategy prioritizes video services while preserving fairness. These results demonstrate the algorithm's ability to adapt resource allocation to heterogeneous service demands. Fig. 3 compares performance versus average transmit power. The proposed method consistently outperforms all baselines. At high power, it eliminates playback interruptions (100% reduction) and significantly reduces overflow probability (about 88%–91% improvement). It also achieves lower delay (25.44 ms), corresponding to a 29.3%–36.9% reduction. Overall, the proposed algorithm effectively balances reliability, overflow control, and delay.

V. Conclusion

We propose a low-complexity DRL framework for QoE-driven scheduling using both CSI and QSI. A hierarchical design with CSIS-Net and QSP-Net efficiently handles large state–action spaces and significantly reduces video interruptions, buffer overflows, and delay in MU-MIMO-OFDM networks.

REFERENCES

[1] U. Kulkarni, K. Diab, L. Cao, F. Ahmed, S. Aggarwal, P. Sharma, and S. Fahmy, "Maestro: Qoe-aware dynamic resource allocation in wi-fi networks," *Proceedings of the ACM on Networking*, vol. 3, no. CoNEXT1, pp. 1–24, 2025.

[2] S. Chai and V. K. N. Lau, "Joint rate and power optimization for multimedia streaming in wireless fading channels via parametric policy gradient," *IEEE Transactions on Signal Processing*, vol. 67, no. 17, pp. 4570–4581, 2019.

[3] V. K. N. Lau and F. Zhang, "Optimal beamforming for video streaming in multiantenna interference networks via diffusion limit," *IEEE Transactions on Information Theory*, vol. 61, no. 4, pp. 1819–1841, 2015.

[4] R. Fritzsche, P. Rost, and G. P. Fettweis, "Robust rate adaptation and proportional fair scheduling with imperfect csi," *IEEE Transactions on Wireless Communications*, vol. 14, no. 8, pp. 4417–4427, 2015.

[5] C. Pan, H. Zhu, N. J. Gomes, and J. Wang, "Joint precoding and rrh selection for user-centric green mimo c-ran," *IEEE Transactions on Wireless Communications*, vol. 16, no. 5, pp. 2891–2906, 2017.

[6] A. Razi, D. J. Ryan, I. B. Collings, and J. Yuan, "Sum rates, rate allocation, and user scheduling for multi-user mimo vector perturbation precoding," *IEEE Transactions on Wireless Communications*, vol. 9, no. 1, pp. 356–365, 2010.

[7] A. A. Khan, R. S. Adve, and W. Yu, "Optimizing downlink resource allocation in multiuser mimo networks via fractional programming and the hungarian algorithm," *IEEE Transactions on Wireless Communications*, vol. 19, no. 8, pp. 5162–5175, 2020.

[8] Y. Xu, G. Yue, N. Prasad, S. Rangarajan, and S. Mao, "User grouping and scheduling for large scale mimo systems with two-stage precoding," in *2014 IEEE International Conference on Communications (ICC)*, Sydney, NSW, Australia, 2014, pp. 5197–5202.

[9] G. Lee and Y. Sung, "A new approach to user scheduling in massive multi-user mimo broadcast channels," *IEEE Transactions on Communications*, vol. 66, no. 4, pp. 1481–1495, 2018.

[10] L. Lei, Y. Kuang, N. Cheng, X. S. Shen, Z. Zhong, and C. Lin, "Delay-optimal dynamic mode selection and resource allocation in device-to-device communications—part i: Optimal policy," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 5, pp. 3474–3490, 2016.

[11] Y. Cui and V. K. N. Lau, "Delay-aware bs-dtx control and user scheduling for energy harvesting downlink coordinated mimo systems," in *2012 IEEE 51st Conference on Decision and Control (CDC)*, Maui, HI, USA, 2012, pp. 5852–5857.

[12] J. Wang, C. Xu, Y. Huangfu, R. Li, Y. Ge, and J. Wang, "Deep reinforcement learning for scheduling in cellular networks," in *2019 11th International Conference on Wireless Communications and Signal Processing (WCSP)*, Xi'an, China, 2019, pp. 1–6.

[13] Q. Lan, B. Lv, R. Wang, K. Huang, and Y. Gong, "Adaptive video streaming for massive mimo networks via approximate mdp and reinforcement learning," *IEEE Transactions on Wireless Communications*, vol. 19, no. 9, pp. 5716–5731, 2020.

[14] M. Cheng, J. Li, and S. Nazarian, "Drl-cloud: Deep reinforcement learning-based resource provisioning and task scheduling for cloud service providers," in *2018 23rd Asia and South Pacific Design Automation Conference (ASP-DAC)*, Jeju, South Korea, 2018, pp. 129–134.

[15] Q. An, S. Segarra, C. Dick, A. Sabharwal, and R. Doost-Mohammady, "A deep reinforcement learning-based resource scheduler for massive mimo networks," *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 1, pp. 242–257, 2023.

[16] X. Guo, Z. Li, P. Liu, R. Yan, Y. Han, X. Hei, and G. Zhong, "A novel user selection massive mimo scheduling algorithm via real time ddpg," in *GLOBECOM 2020 - 2020 IEEE Global Communications Conference*, Taipei, Taiwan, 2020, pp. 1–6.

[17] L. Kleinrock, *Queueing Systems, Volume 1: Theory*, 1st ed. New York: Wiley Interscience, 1975.